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Build what next India runs on

Team Name : Limit Breakers

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Problem Statement : Traditional hiring systems rely heavily on keyword matching, causing recruiters to miss highly relevant candidates.

Solution Overview

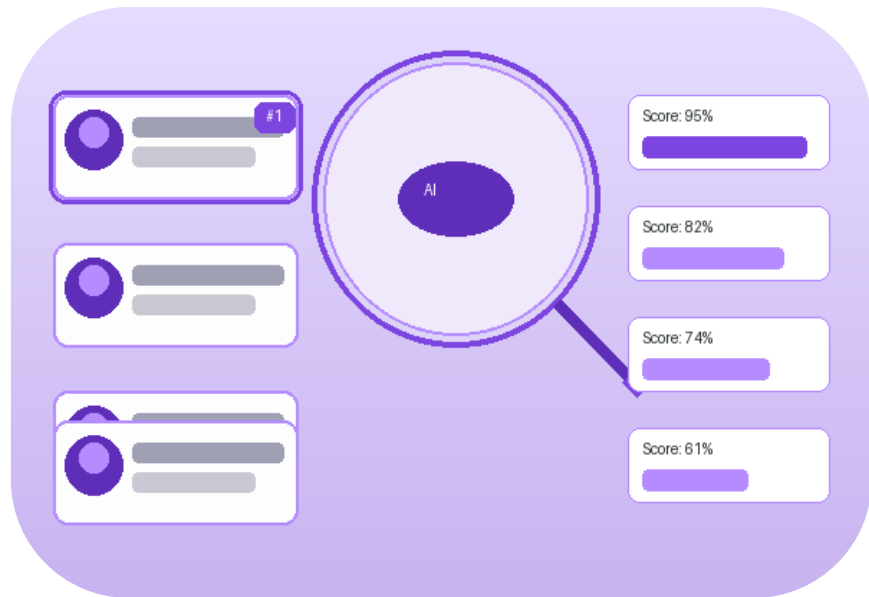
What is your proposed solution?

An explainable AI-powered candidate ranking system. Our system evaluates candidates using:

- **What they know** (skills & experience)
- **What they have done** (career evidence)
- **How they behave** (activity & recruiter signals)
- **Why they fit the role** (explainable ranking)

What differentiates your approach?

- Goes beyond keyword matching
- Uses behavioral hiring signals
- Detects suspicious profiles and honeypots
- Provides recruiter-friendly explanations
- Produces trusted, explainable rankings



JD Understanding & Candidate Evaluation

Key JD Requirements

- Search & Retrieval Systems • Ranking Systems • Recommendation Engines
- Production ML Experience • Python Expertise • Product Mindset

Most Important Candidate Signals

Technical: Retrieval/Search Experience, Ranking Systems, Recommendation Systems, Production Deployments

Behavioral: Recruiter Response Rate, Activity Recency, Interview Completion, Open-to-Work Status

Career: Ownership, Startup Experience, Product Impact

Rather than matching keywords, the system evaluates whether a candidate has demonstrated the capabilities required for the role.



Ranking Methodology

Ranking Pipeline

Parse JD requirements → Extract candidate features →
Compute semantic relevance → Score behavioral
signals → Score career evidence → Detect honeypots
→ Generate final ranking

Models, Algorithms & Heuristics

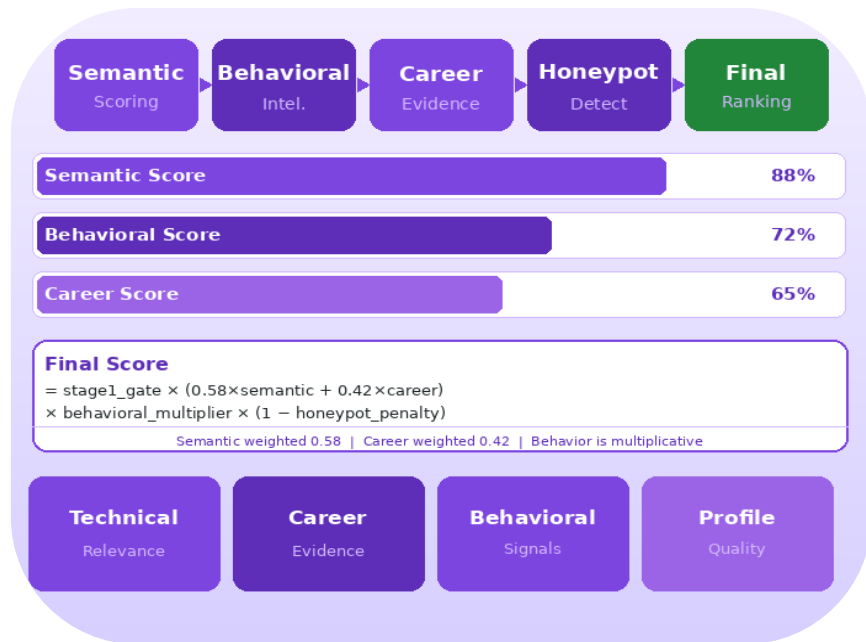
Semantic Similarity Scoring • Feature Engineering • Multi-
Stage Ranking Pipeline • Behavioral Signal Weighting •
Honeypot Detection Rules

Final Score Formula

Final Score = stage1_gate × (0.58 × semantic + 0.42 ×
career_evidence) × behavioral_multiplier × (1 –
honeypot_penalty)

The ranking score combines:

- **Technical relevance** – does the candidate know the subject?
- **Career evidence** – have they done it before?
- **Behavioral signals** – are they actively engaged and reachable?
- **Profile quality checks** – is the profile credible and consistent?



Explainability & Data Validation

How Ranking Decisions Are Explained

Each candidate receives: Overall Score • Positive Factors
• Negative Factors • Skill Alignment • Career Highlights • Behavioral Highlights

Preventing Hallucinations

Evidence-based explanations • Only profile-derived insights • No unsupported generated claims

Handling Suspicious Profiles

Timeline Validation • Contradiction Detection •
Keyword Stuffing Checks • Honeypot Detection •
Risk Penalties

Every recommendation includes both strengths and risk factors so recruiters can understand why a candidate was ranked.

Sample Explainability Output

Candidate #18472 | Final Score: 94.2

Top Strengths

- ✓ Retrieval Systems experience
- ✓ Ranking Infrastructure background
- ✓ Product Experience (2 shipped products)
- ✓ High Recruiter Engagement score

Risk Factors

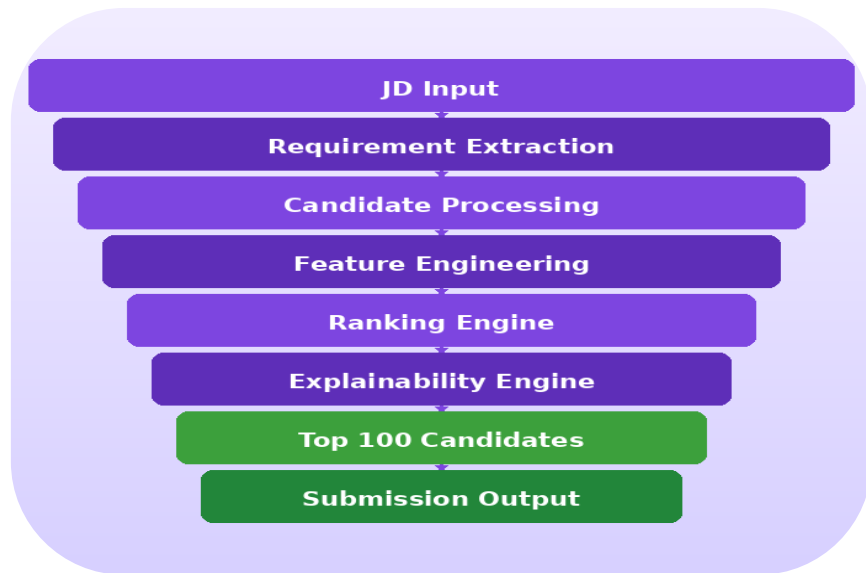
⚠ Longer notice period (60 days)

Every ranking decision is fully explainable.

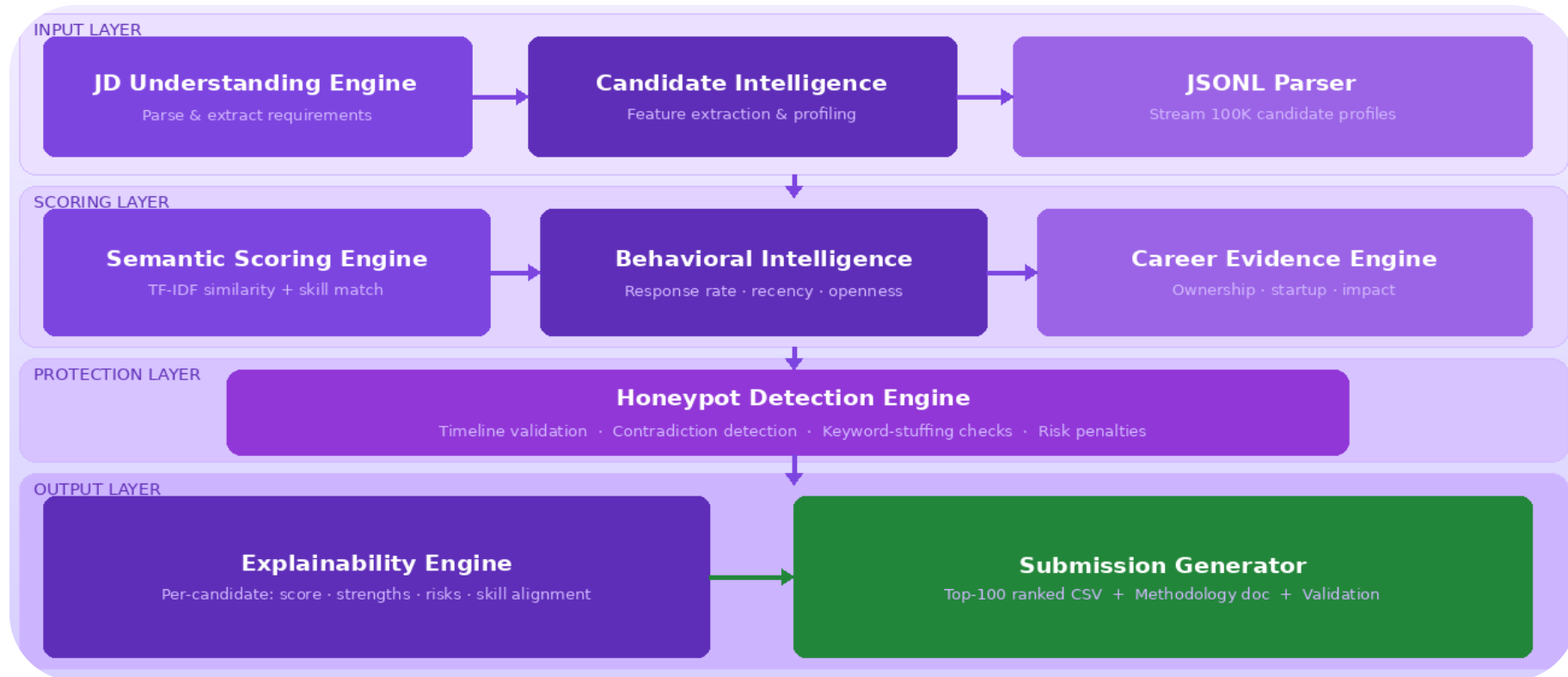
End-to-End Workflow

Complete Workflow: JD Input to Ranked Output

- ① JD Input → ② Requirement Extraction → ③
Candidate Processing
④ Feature Engineering → ⑤ Ranking Engine → ⑥
Explainability Engine
⑦ Top 100 Candidates → ⑧ Submission Output



System Architecture



The architecture separates candidate understanding, scoring, validation, and explainability into independent modules for scalability and transparency.

Results & Performance

Ranking Quality Insights

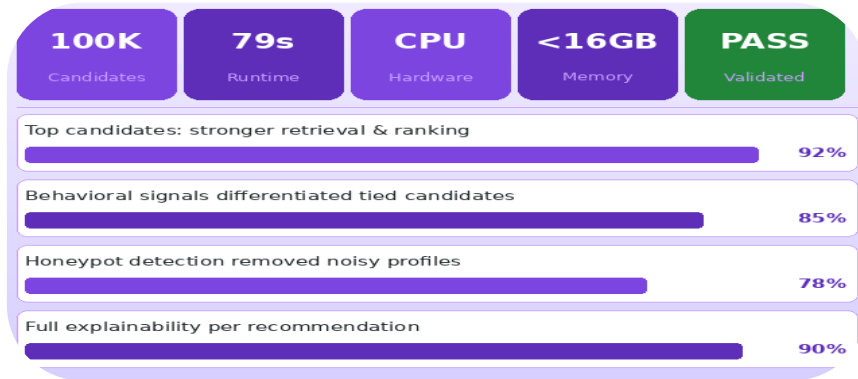
- Top-ranked candidates consistently showed stronger retrieval, ranking, and recommendation-system experience than lower-ranked ones.
- Behavioral signals helped distinguish technically similar candidates where skill overlap alone was insufficient.
- Honeypot detection removed suspicious profiles from the final shortlist, reducing noise for recruiters.
- Explainability enabled transparent recruiter review — every recommendation included positive factors, risks, and skill alignment.

Runtime & Constraints

Metric	Result
Candidates Processed	100,000
Runtime	79 seconds
Hardware	CPU Only
Memory Usage	<16 GB
Submission Validation	Passed 

Business Impact

- Reduces dependence on keyword matching
- Surfaces hidden but relevant candidates
- Improves recruiter trust through explainability
- Filters suspicious profiles automatically



Key Outcomes

-  100K candidates processed in 79 sec on CPU
-  Explainable rankings for every candidate
-  Suspicious profiles filtered — honeypots removed
-  Validation-compliant submission passed

Technologies Used

Technology Why We Used It

- **Python** Rapid AI system development; rich ecosystem for NLP and data pipelines
- **Pandas** Efficient structured processing of 100,000 candidate profiles at scale
- **NumPy** Fast vectorised numerical scoring across all candidate signals
- **JSONL Processing** Streaming candidate parsing without loading all 100K records into memory
- **Modular Pipeline** Each engine is independently testable – supports explainability and maintainability

No external ML APIs. No cloud dependencies. Fully self-contained and reproducible.

Design Principles

- ✓ No External ML APIs
- ✓ Fully Offline Execution
- ✓ Reproducible Results
- ✓ CPU-Only Processing
- ✓ Explainability First

Submission Assets

Deliverables

- ✓ GitHub Repository
- ✓ Ranked Candidates CSV
- ✓ Explainability Report
- ✓ PPT Submission
- ✓ Validation-Compliant Output

Repository

github.com/pavankarthikeyaatchyuta-lab/Redrob-Intelligent-Candidate-Ranker

“We didn’t build a keyword matcher. We built an explainable AI recruiter that understands candidate relevance, hiring readiness, and recruiter intent.”



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THANK YOU